MAE301, Group 24
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Phase 3 Report

Executive Summary:

Our MVP addresses a growing challenge faced by independent investors who want to make financially informed decisions while staying true to their personal ethics. Many investors are looking to grow their portfolios but may feel conflicted about supporting corporations whose practices do not align with their personal values. To address this issue, we've developed an AI agent that analyzes companies using both financial data and ethical (ESG) metrics, then recommends alternative investments within similar sectors that better reflect user preferences. Users can input a company name or ticker along with their ethical preferences for investment, then our system returns an ethical analysis, financial summary, and alternative stock recommendation that align with both performance and values.

User and Use Case:

The expected user for this MVP is an independent or beginner investor who wants to gain a better understanding of the stock market and invest in it while making conscious decisions about the various aspects they believe make a company ethical. This system may be used as a form of maintaining a connection to improve an understanding of real-life ESG ratings, including factors such as employee treatment, environmental interest, and which sectors they are investing for in the future. For example, a user considering investing in a large corporation can enter the company name or ticker and provide ethical preferences related to environmental impact, employee treatment, data privacy, corporate governance, and transparency. The agent then analyzes the companies financial and ESG information and returns a scale on how well the company matches the users priorities. It then can also suggest alternative companies in similar sectors that may align better with the user while maintaining financial stability.

System Design

Data:

The system combines financial, ESG, and public perception data from sources such as Yahoo Finance and Alpha Vantage to acquire stock prices and trading trends, SEC EDGAR filings for financial reports, Kaggle ESG datasets and company sustainability reports to get a better understanding for ethical metrics, and also a potential addition of news sources like NewsAPI or GDELT to get a public's general thoughts on a company. The MVP uses a limited dataset of roughly 100–500 companies, along with some ESG coverage. Data is cleaned by standardizing financial units, normalizing ESG scores, and removing unrelated and/or incomplete entries. Datasets are aligned by company and timeframe and handle missing values through filtering.

Models:

ChatGPT and Codex used for building MVP through prompt feed and optimization. We broke the main objectives of the project into small steps in order to ensure each part of the project worked separately, then put everything together after troubleshooting each phase and

optimizing them individually. For the prompting used in individual phases, we wrote clear and distinguishable steps to pull the exact information required, then worked with ChatGPT to improve the prompting that we would feed to Codex.

Evaluation

The system is evaluated through both quantitative and qualitative methods. Quantitatively, we compare the financial performance of recommended stocks to original selections using return differences, correlation coefficients, and volatility measures, while also validating credibility scores against available ESG ratings and tracking system coverage and confidence calibration. Qualitatively, we look at performances through user scenario testing, manual reviews of outputs, and error analysis to identify issues, missing data being a primary example.

Limitations and Risks:

Some limitations in our system stem from restricted access to AI development tools and data resources. While we utilized ChatGPT and Codex to build our agent, we were unable to surpass usage limits for further experimentation. Additionally, access to high quality financial and ESG data was limited and many reliable sources required paid APIs, which reduced our data pool and impacted performance. Some other risks include reliability issues of the agent where the agent may produce incorrect or unsupported conclusions. There is also a financial risk that the agent may recommend alternative investments that underperform, even if they better align with a user's ethical preferences.

Next Steps:

With an added timeline of 2-3 more months, we would focus on improving the connection between stock performance data, ESG data, and user specified preferences. This includes expanding the number of companies in our dataset and improving the accuracy of comparable stock recommendations. This would also include allowing the user to choose what data they are particularly interested in, and having the program not show other ethical/stock information that the customer finds irrelevant. Furthermore, we would focus heavily on improving the interface so that the system is easier to navigate and more accessible for beginner investors. Finally, we would add more testing through example scenarios that allow us to analyze common error and outputs to determine where the agent performs well and where we need to improve.